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PRELIMINARY PALEOMAGNETIC RESULTS ON SOME TRIASSIC ROCKS FROM EAST GREENLAND

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Paleomagnetic measurements have been carried out on six samples of Early Triassic age and five samples of Middle Triassic age from East Greenland. The mean stable remanent magnetization directions obtained after alternating-field demagnetization tests give the virtual geomagnetic pole positions as: (1) 49°N, 158°E for the Middle Triassic and (2) 34°30'N, 176°W for the Early Triassic. The Greenland Triassic paleomagnetic results have been compared with those for Europe and North America. It is inferred from this comparison that these preliminary results for Greenland do not conform with the requirements of a reconstruction based on a geometrical fit of the three landmasses.

1. Introduction

Paleomagnetic results on Early Tertiary lavas from East Greenland (Tarling, 1967) and West Greenland (Tarling and Otulana, 1972; Kristjansson and Deutsch, 1973; Athavale and Sharma, in press) have provided information about the drift of Greenland during the Tertiary period. Bidgood and Harland (1961) reported paleomagnetic measurements on Permian and Triassic rocks from East Greenland, and these were the only available data prior to the present study for determining the position of Greenland in a pre-drift configuration of the landmasses surrounding the North Atlantic Ocean. These results were, however, based on measurement of natural remanent magnetization directions only, and no stability tests involving alternating-field demagnetization or thermal demagnetization had been applied. It was therefore not possible to ascertain whether the directions reported by them represented the directions of primary remanent magnetization in the Permian and Triassic rocks of Greenland.

We report in this paper the results of paleomagnetic measurements on a few samples collected from two Triassic horizons in East Greenland. Alternating-field demagnetization tests on these samples showed that they possessed stable remanent magnetization. However, we treat these results as preliminary, mainly because of the small number of samples involved in our study.

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2. Sampling details and measurement procedures

In total, eleven samples of fine-grained red coloured sandstone collected from the Scoresby Sund–Jameson Land area (Fig. 1) of East Greenland were processed by us. Six oriented samples collected from the Rødstaken member horizon of the Pingodal Formation and belonging to the Scythian stage (Early Triassic) were supplied to us by Prof. K. Birkenmajer. Five oriented samples collected from the Solfaldsdal horizon in the Gipsdalen Formation and belonging to the Anisian stage (Middle Triassic) were supplied by Dr. K. Perch-Nielsen. A general description of the Triassic rocks from East-Greenland has been given by Trümpy (1961). Detailed description of the geological setting, lithology and age of rocks from the area of sampling has been given by Perch-Nielsen et al. (1972).

Four specimens were obtained from each sample and the directions of natural remanent magnetization (NMR) were measured on a PAR Spinner-magnetometer. One specimen from each sample was subjected to alternating-field demagnetization in successively increasing peak-field strengths up to a maximum value of 2800 Oersteds. The response of two typical specimens, one each from the Rødstaken and the Solfaldsdal group of samples, to alternating-field treatment

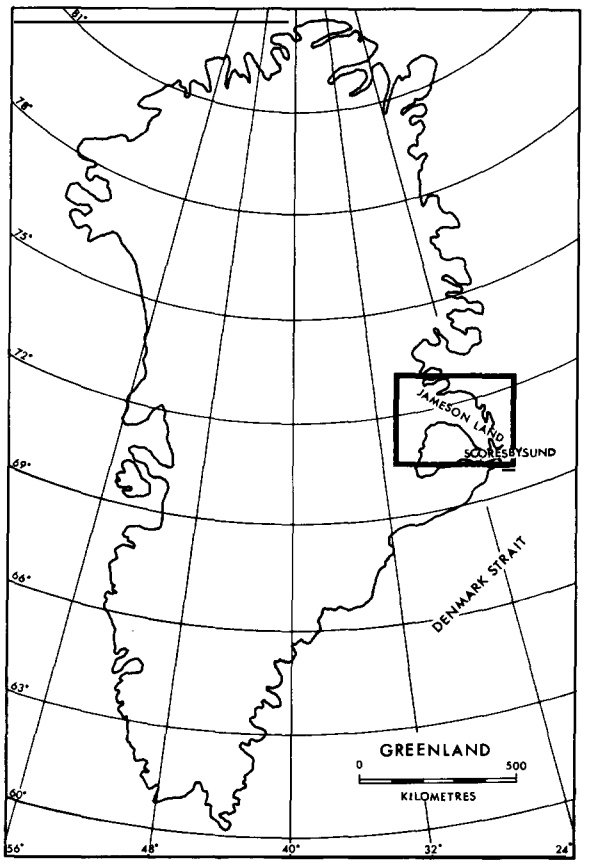


Fig. 1. Approximate location of sampling area.

is illustrated in Fig. 2A and 2B. The pilot specimens were found to possess high coercivity and did not show relatively big loss of remanent magnetic intensity on demagnetization in peak fields upto 2800 Oe (Fig. 2A). The directions of remanent magnetization obtained after demagnetization in successively increasing peak-field values were, in general, found to remain well grouped up to the peak value of 1000 Oe and to disperse increasingly for higher values of the demagnetizing field (Fig. 2B). The peak-field value of 1000 Oe was therefore chosen for batch cleaning of the remaining specimens.

The cleaned directions for specimens from the two horizons were found to fall in two distinct groups. Mean stable remanent direction of magnetization for each group of samples was obtained by using the statistical method described by Fisher (1953). The specimens were given unit weight in obtaining the mean direction for a sample and the mean directions of samples were given unit weight for obtaining the mean direction for each group. The horizons sampled for our study have very gentle structural dips (0°–5°), and tilt corrections were not applied to the mean directions.

Both the sample groups were found to be normally magnetized. The mean directions and the virtual geomagnetic poles calculated from them are presented in Table I along with other relevant information.

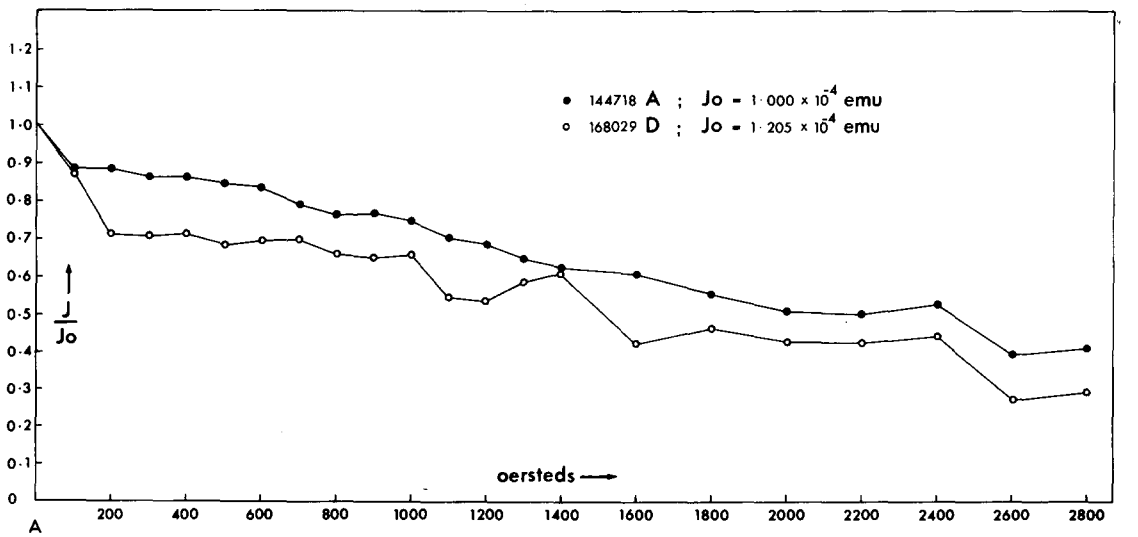


Fig. 2A. Variation of normalized magnetic intensity J/J_0 with peak alternating-field strength for one specimen each from the Rødstaken (•) and Solfaldsdal (○) groups of samples.

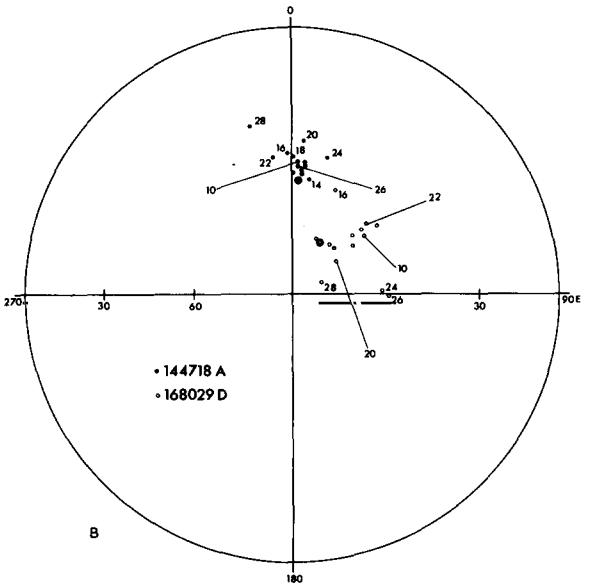


Fig. 2B. Change in the direction of magnetization of the two specimens illustrated in Fig. 2A. The values of demagnetizing field strength for some of the measured directions are indicated in units of hundreds of Oersteds. The NRM directions are indicated by an enclosing circle. Lambert equal-area projection.

3. Analysis of Triassic paleomagnetic data from Europe and North America

The virtual geomagnetic pole positions obtained from rocks of Early and Middle Triassic ages from Greenland are plotted in Fig. 3A. In order to compare these results with those for Triassic rocks from North America and Europe, we have analyzed the paleomagnetic data available in the latest compilation by McElhinny (1973). Additional information about some of the results was obtained from the earlier compilation by Irving (1964) and from the lists of paleomagnetic pole positions published periodically by McElhinny in the Geophysical Journal of the Royal Astronomical Society.

McElhinny (1973) has classified the European paleomagnetic data into three sub-groups: (1) Western Europe; (2) Russian platform region west of the Ural Mountains; and (3) Southern European region south of the Alpine front. He has suggested that the first two sub-groups represent a single tectonic unit for the purpose of comparison of European paleomagnetic data with those for the other landmasses. The Southern European results are, however, considered circumspect for this specific purpose, in view of the tectonic disturbances to which this region had been subjected during the Tertiary period. Roy (1972) has expressed a similar view.

MacElhinny (1973, p. 288) has listed four paleomagnetic poles for Western Europe (No. WE 8.1 to WE 8.4) from Triassic rocks. Of these, the pole for Ladinian volcanics (WE 8.4) has to be omitted for two reasons. First, this formation is located in northern Italy (coordinates 46.4°N, 11.7°E) and should therefore have been grouped with the other Southern European results. Secondly, the statistical errors on

TABLE I

Paleomagnetic directions for Triassic rocks from East Greenland

Formation	Age	Locality coordinates	No. of samples	Mean stable remanent magnetization direction		α_{95}	Virtual geomagnetic pole			
				D	I		lat.	long.	δp	δm
Rødstaken member of Pingodal Formation	Early Triassic (Scythian)	72°N 24°W	6	336	+33	6.8°	34°30'N	176°W	4.4°	7.7°
Solfaldsdal member, Gipsdalen Formation	Middle Triassic (Anisian?)	72°N 23°W	5	001	+62	9.9°	49°N	158°E	12°	15°

D = declination, I = inclination, α_{95} = radius of circle of confidence, δp, δm = semi-axes of oval of confidence at 95% level.

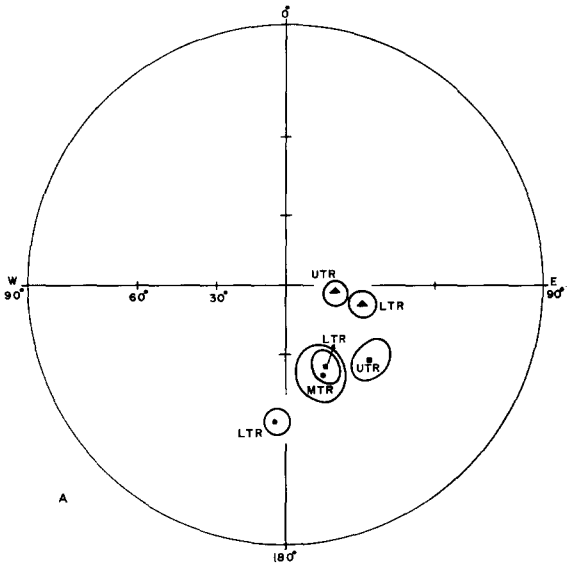


Fig. 3A. Triassic (*TR*) paleomagnetic pole positions for Greenland (•), Europe (■) and North America (▲). Prefixes *L*, *M* and *U* have been used to indicate the Lower, Middle and Upper Triassic divisions, respectively. Equatorial stereographic projection.

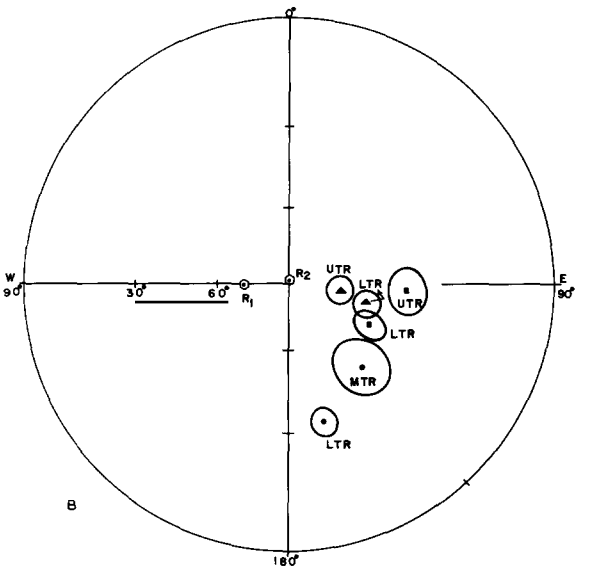


Fig. 3B. New locations of Triassic paleomagnetic poles for Greenland and Europe on displacement of these landmasses according to the Bullard et al. (1965) model. Equatorial stereographic projection. R_1 = Rotation pole for Greenland. R_2 = Rotation pole for Europe.

this result are not available (McElhinny, 1969). Pole position WE 8.3 for Keuper marls from Sidmouth, England, is also listed without any statistical errors (Irving, 1964). Pole position WE 8.1 is the mean pole position for Vosges sandstones from France. The only criterion for stability of the magnetization measured in this case is the occurrence of normally and reversely magnetized rocks. However, the mean directions for normal and reverse groups are far from being antiparallel and are thus suggestive of the presence of secondary components of magnetization. The only remaining result (WE 8.2) for Triassic from Western Europe is therefore a separate study of Upper Triassic Keuper marls from England (Irving, 1964).

The Triassic paleomagnetic data from the Russian platform region (No. RP 8.1 to RP 8.10, in McElhinny, 1973, p. 294) appear to have been obtained after direct or alternating-field demagnetization tests. Nine out of the ten pole positions are, however, based on samples of Early Triassic age and only one pole position is for rocks of Middle or Late Triassic age. We have, therefore, used the mean pole position given by McElhinny (1973) for the Russian platform as representative of Early Triassic, and the single pole for Keuper marl from England as representative of Late Triassic for Europe, for comparison with paleomagnetic results from North America and Greenland.

The Triassic paleomagnetic data from North

TABLE II

Triassic Paleomagnetic data from Europe and North America

Period	Europe					North America		
	lat.	long.	δp	δm	α_{95}	lat.	long.	α_{95}
Late Triassic	43°N	131°E	7°	13°	–	68°N	97°E	6°
Early Triassic	51°N	154°E	–	–	7°	56°N	104°E	6°

δp; δm = radii of semi-axes of oval of confidence, α_{95} = radius of circle of confidence.

America fall into two different groups of Early and Late Triassic (McElhinny, 1973). The mean pole positions are based on results from several formations, for which alternating-field and/or thermal demagnetization tests had been applied.

The various Triassic pole positions for North America and Europe are listed in Table II and plotted in Fig. 3A.

4. Discussion of results

Comparisons of paleomagnetic data from Europe and North America have been carried out by several workers (Wells and Verhoogen, 1967; Hospers and Van Andel, 1968; Deutsch, 1969; Creer, 1970; Larson and LaFountain, 1970; Phillips and Forsyth, 1972). These studies showed, in general, that the paleomagnetic pole positions for these landmasses converge together for the period Cambrian to Triassic, when one of the landmasses is rotated with respect to the other according to the morphological reconstruction model suggested by Bullard et al. (1965). The polar wandering paths of the two continents match with each other up to Triassic and then start diverging, with the separation increasing during Jurassic and Cretaceous. The paleomagnetic data are, therefore, considered as indicative of a separation of the two landmasses and opening of the North Atlantic sometime during the Triassic. On the basis of analysis of paleomagnetic data in terms of paleolatitudes, Roy (1972) has suggested that the rupture between the two landmasses took place during Early Triassic.

The quiet zone of magnetic anomalies, which extends from the eastern shelf margin of the United States up to a distance of about 400 km in the North Atlantic, is considered by Pitman and Talwani (1972) to be of Triassic to Jurassic age. Vogt and others (1970) have suggested that the low amplitude of magnetic anomalies in the quiet zone is due to generation of the oceanic crust along a N–S trending ridge at equatorial altitudes during the Early Triassic. The interpretation of Paleozoic and Mesozoic paleomagnetic data from North America and Europe as well as the marine magnetic anomaly data for the North Atlantic therefore suggests that separation of the two landmasses started during Early Triassic.

Recent seismic studies in the Baffin Bay separating northeast Canada From West Greenland show that it has an oceanic crust and the thickness of sediments deposited on this crust is up to 7 km (Keen and Barret, 1972). Sediments of comparable thickness are found on the shelf margins on both sides and Henderson (1973) has suggested on the basis of detailed geological–geophysical studies of the West Greenland shelf that the offshore sediments could be Jurassic or even older in age. Dawes (1973) has carried out a detailed investigation of the North Greenland fold belt and shown that direction of Paleozoic compression in it is opposite to that of contemporaneous compression in the Ellesmere Island of Canada. He has concluded from these studies that the Nares Strait lineament, which connects Baffin Bay with the Arctic basin, could be of Paleozoic (pre-Carboniferous) age and has suggested that the Nares Strait, Baffin Bay and Davis Strait could be the products of the same episode of continental break-up. These observations suggest that Greenland had separated from North America during or even before the Triassic period. The separation between Greenland and Europe is considered to have taken place much later in Late Cretaceous times, with the opening of the Northeast Atlantic Ocean (Pitman and Talwani, 1972).

The reconstruction model proposed by Bullard et al. (1965), involves a geometrical fit between the three landmasses of North America, Europe and Greenland, at their margin represented by the 500-fathom contour line. In this reconstruction, they have first rotated Greenland anticlockwise to fit with Europe and the Europe–Greenland block is then rotated clockwise through 38°, around a pole of rotation located at

TABLE III

Displaced pole positions for Europe and Greenland (Triassic) on rotation according to Bullard et al. (1965) model

Period	Latitude (°N)	Longitude (°E)
Greenland		
Early Triassic	32	165
Middle Triassic	45	138
Europe		
Early Triassic	52	116
Upper Triassic	42	94

88.5°N, 27.7°E, to fit with North America. The resultant movement of Greenland amounts to a clockwise rotation through 18° around a rotation pole located at 70.5°N and 94.4°W. (Wells and Verhoogen, 1967).

Deutsch (1969) has given a method for calculating the displacement of paleomagnetic pole positions as a consequence of movements of landmasses. We have used this method in locating the new positions for the Triassic paleomagnetic poles for Greenland and Europe (Fig. 3B) when these landmasses are rotated to fit with North America according to the Bullard et al. (1965) model.

In Fig. 3B, the pole positions for Europe and North America show an overlap, indicating a general agreement of the Triassic paleomagnetic results with the Bullard et al. (1965) reconstruction model.

The Early Triassic pole from Greenland is, however, located far off from this cluster of European and North American poles. The Middle Triassic Greenland pole falls close to this cluster. However, this result is associated with a large error. Assuming that the preliminary results presented here are representative of the Triassic geomagnetic field in Greenland, we conclude that they do not conform with the requirements of the reconstruction model proposed by Bullard et al. (1965), and suggest that the drift history of Greenland may be different from that assumed for constructing a simplistic geometric fit. The paleomagnetic results obtained for the Early Tertiary lava sequences from West Greenland (Athavale and Sharma, in press) have led us to furmulate a similar interpretation regarding the movement of Greenland.

Alternatively, the nonconformity of the Triassic Greenland data with the Bullard et al. (1965) model could be due to the possibility that the Triassic geomagnetic field was not geocentric, and had substantial nondipole components superimposed on the dipole field, as has been suggested by Briden et al. (1971) from an analysis of global Triassic paleomagnetic data.

Additional paleomagnetic data from Greenland for the Paleozoic and Mesozoic periods are required for examining the veracity of these conclusions.

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